DELHI PUBLIC SCHOOL JANKIPURAM, LUCKNOW

SESSION 2025-2026: PERIODIC ASSESSMENT-I

SET-I

Subject: Artificial Intelligence (417)

TIME: $1\frac{1}{2}$ Hrs. M.M.:20

Date: _____ Name: _____ Class/ Sec X_____

General Instructions:

Read the following instructions carefully and follow them:

- This Question Paper consists of 21 questions in two sections: Section A & Section B.
- Section A has Objective type questions whereas Section B contains Subjective type questions.
- Out of the given (12 + 9 =) 21 questions, a candidate has to answer (8 + 5 =) 13 questions in the allotted (maximum) time.
- All questions of a particular section must be attempted in the correct order.
- Section A - Objective type questions (11 marks):
- Section B – Subjective type questions (14 marks):

SECTION A: OBJECTIVE TYPE QUESTIONS

•	**Answer any 2 out of the given 4 questions on Employability Skills (1 x 2 = 2 marks)**	
i.	_____ is the response given with respect to the message, received by the receiver. (a) Encoding (b) Response (c) Feedback (d) Both (a) and (b)	1
ii.	Assertion (A): Feedback is not necessary in the communication process. Reason (R): Once the message is sent, the process is considered complete. Select the appropriate option for the statements given above: Option (a): Both (A) and (R) are true and (R) is the correct explanation of (A). Option (b): Both (A) and (R) are true but (R) is not the correct explanation of (A). Option (c): (A) is true but (R) is false. Option (d): (A) is false but (R) is true.	1
iii.	_____ means being accountable for your tasks and actions. (a) Self awareness (b) Responsibility (c) Adaptability (d) Time management	1
iv.	Statement 1: Stress related to money or unplanned savings and expenditure of income is called as financial stress. Statement 2: Financial stress arises due to lack of money, worried about more expenses or don't have money to buy comforts or luxuries of life. a. Both Statement 1 and Statement 2 are correct b. Both Statement 1 and Statement 2 are incorrect c. Statement 1 is correct but Statement 2 is incorrect d. Statement 2 is correct but Statement 1 is incorrect	1
•	**Answer any 6 out of the given 8 questions (1 x 6 = 6 marks)**	
v.	Data exploration stage involves_____of large volumes of data to uncover meaningful patterns using charts, plots, graphs etc. (a) Activation (b) Detection (c) Analysis (d) Modification	1

vi.	In 4W's problem canvas _____ block identifies the specific problem or challenge that needs to be addressed. (a) Who? (b) What? (c) Where? (d) Why?	1
vii.	Google lens is an application of Google that can identify objects in images by analysing a photo from the real world. (True/False)	1
viii.	**Statement 1:** Computer vision is widely used in surveillance system to enhance security in public spaces, buildings and border areas. **Statement 2:** Self – driving cars utilise CV to perform aerial inspections, monitor large areas or assess damage after natural disaster. a. Both Statement 1 and Statement 2 are correct b. Both Statement 1 and Statement 2 are incorrect c. Statement 1 is correct but Statement 2 is incorrect d. Statement 2 is correct but Statement 1 is incorrect	1
ix.	_____ is one of the most widely used techniques for building AI system worldwide. (a) Artificial Inteligence (b) Machine Learning (c) Deep Learning (d) Artificial Engineering	1
x.	The platforms like Netflix, Flipkart and Spotify etc use _____ to help their customers to find the related products. (a) Recommendation System (b) Spam Email Filtering (c) Image Recognition (d) Anamoly Detection	1
xi.	ANN uses _____ algorithm that takes input image, process it by assigning learnable weights and biases to objects. a) Machine Learning (b) Deep Learning (c) Accomodated Engineering (d) None of these	1
xii.	**Assertion (A):** Unlabelled data is used in supervised learning. **Reason (R):** Supervised learning needs data labelled data to find hidden patterns. Select the appropriate option for the statements given above: **Option (a):** Both (A) and (R) are true and (R) is the correct explanation of (A). **Option (b):** Both (A) and (R) are true but (R) is not the correct explanation of (A). **Option (c):** (A) is true but (R) is false. **Option (d):** (A) is false but (R) is true.	1

SECTION B: SUBJECTIVE TYPE QUESTIONS

	Answer any 1 out of the given 2 questions on Employability Skills (1 x 2 = 2 marks) **Answer each question in 20 – 30 words.**	
xiii.	What do you understand by the term strength? How it is different from weakness?	2
xiv.	What do you understand by self – regulation? Using two points tell how it helps us?	2
•	**Answer any 3 out of the given 5 questions in 20-30 words each. (2 x 3 = 6 marks)**	
xv.	Discuss Privacy risk and Autonomous weapon as the major challenges related to AI.	2
xvi.	What do you understand by bioethics in sector based ethical frameworks?	2
xvii.	What do you understand by training data set? How it is different from testing data set?	2
xviii.	What are labels? Explain with example.	2
xix.	What is the role of anamoly detection in medical diagnosis?	2
•	**Answer any 1 out of the given 2 questions in 50– 80 words each (4 x 1 = 4 marks)**	
xx.	Write and discuss any four applications of Computer vision with proper headings.	4
xxi	Why do we need ethical framework for AI? Explain through any four points with proper headings.	4